

Quantitative Research Methods

Chapter 10: Deep Learning

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The course at a glance

Lecture	Topic
1	Ch. 1 + 2 — Introduction; what is statistical learning
2	Ch. 2 — Accuracy, bias–variance, Bayes classifier, KNN
3–4	Ch. 3 — Linear regression (estimation, inference, diagnostics)
5–6	Ch. 4 — Classification (logistic, LDA/QDA, naive Bayes, ROC)
7	Ch. 5 — Resampling (validation, CV, bootstrap)
8	Ch. 6 — Model selection and regularisation (ridge, lasso)
9	Ch. 7 — Moving beyond linearity (splines, GAMs)
10	Ch. 8 — Tree-based methods (trees, forests, boosting)
► 11	Ch. 10 — Deep learning (MLPs, CNNs, training)
12	Ch. 13 — Multiple testing (FWER, FDR)

Twelve sessions of 180 minutes. ► marks today's chapter. Short exercises every ~20 min; extended exercises every ~45 min; each chapter has a companion lab notebook.

Contents

- 1 Why now?
- 2 Single Layer
- 3 Multilayer
- 4 CNNs
- 5 Sequences
- 6 Fitting
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Notation in this chapter

Symbol	Meaning
X_j, p	input feature j ; number of input features
K	number of hidden units in a layer
$g(\cdot)$	activation function (sigmoid, tanh, ReLU)
A_k	activation of hidden unit k : $A_k = g(w_{k0} + \sum_j w_{kj} X_j)$
w_{kj}, w_{k0}	weight from input j into hidden unit k ; its bias
β_k, β_0	output weight of hidden unit k ; output bias
softmax	output activation $e^{z_m} / \sum_{m'} e^{z_{m'}}$ for multiclass labels
W, F, P, S	conv arithmetic: input width, filter size, padding, stride
θ	all weights and biases of the network collectively
η	learning rate of (stochastic) gradient descent
ℓ	per-observation loss (squared error, cross-entropy)

Sources and references

Primary textbook

James et al. (2023). *An Introduction to Statistical Learning, with Applications in Python*. Springer.

Learning objectives

By the end of this chapter you will be able to:

- **Define** a single-layer and multilayer perceptron.
- **Explain** how convolutional networks exploit local structure and translation invariance.
- **Outline** recurrent networks and the basic transformer idea.
- **Train** a small network with stochastic gradient descent + backpropagation.
- **Reason** about when deep learning is worth it.

Roadmap of this chapter

A tour of modern deep learning

- ① Single-layer networks (logistic regression in disguise).
- ② Multilayer perceptrons.
- ③ Convolutional networks for images.
- ④ Recurrent networks and transformers for sequences.
- ⑤ How they are trained (SGD, backprop) and regularised.

Outline

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- 2 Single Layer
- 3 Multilayer
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What is deep learning?

A renaissance of neural networks driven by GPUs, large labelled datasets, and algorithmic improvements (dropout, batch norm, attention).

Takeaway

Dominant in image, speech, and natural-language tasks. Less consistently dominant on small to medium tabular data — where gradient-boosted trees often win.

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Single hidden layer

$$\hat{f}(X) = \beta_0 + \sum_{k=1}^K \beta_k g \left(w_{k0} + \sum_{j=1}^p w_{kj} X_j \right).$$

Reading the formula symbol by symbol

- X_j : input feature j ($j = 1, \dots, p$); g : nonlinear activation (sigmoid, tanh, ReLU).
- w_{kj} : weight from input j into hidden unit k ; w_{k0} : its bias.
- K : number of hidden units (activations $A_k = g(\cdot)$); β_k : unit k 's output weight, β_0 the output bias.

Takeaway

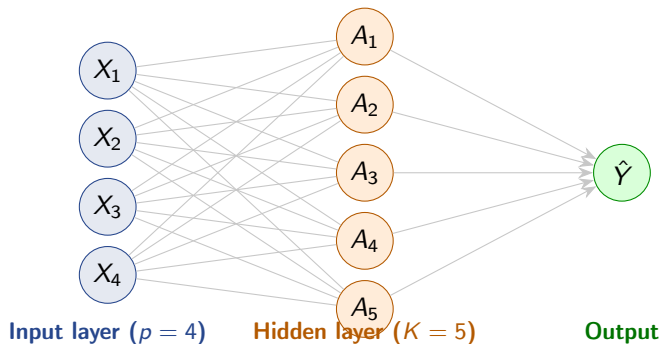
Read it inside-out: linear combination $\rightarrow$ nonlinearity $g \rightarrow$ linear combination. Logistic regression = this with $K = 1$ and a sigmoid output.

Single-layer network as a graph



Source: Figure 10.1 — ISLP, James et al. (2023).

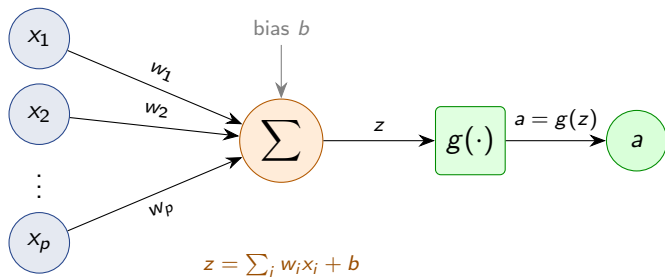
Feed-forward network as a labelled graph



Takeaway

Each hidden unit computes $A_k = g\left(w_{k0} + \sum_j w_{kj}X_j\right)$; the output linearly combines them, $\hat{Y} = \beta_0 + \sum_k \beta_k A_k$. “Fully connected” means every input feeds every hidden unit.

Anatomy of a single neuron



Takeaway

A neuron takes a weighted sum of its inputs plus a bias, then applies a nonlinearity g . Composing layers of these units builds the network above.

Activation functions: sigmoid vs. ReLU

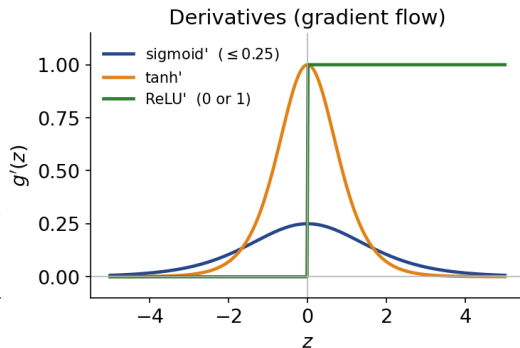
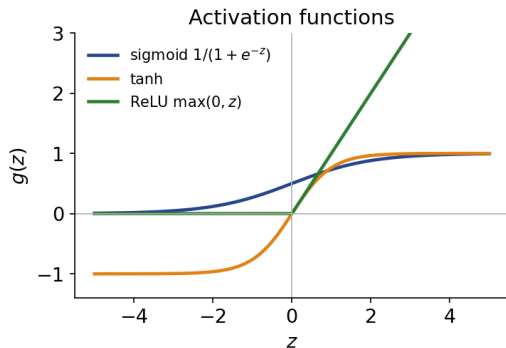
How to read this

- **Sigmoid:** $g(z) = 1/(1 + e^{-z})$, output in $(0, 1)$. Saturates at $\pm\infty$ — vanishing gradients.
- **Tanh:** $g(z) = \tanh(z) = \frac{e^z - e^{-z}}{e^z + e^{-z}}$, output in $(-1, 1)$; same S-shape but zero-centred.
- **ReLU:** $g(z) = \max(0, z)$. Today's default — sparser activations, faster training, better gradient flow.

Why it matters

The activation is the *only* nonlinearity in the network: remove it and stacked layers collapse to one linear map (Exercise 10.2). ReLU is preferred because its gradient stays healthy through deep stacks.

Activation functions and their gradients



Takeaway

ReLU's gradient is 1 on the active region, so signals pass undiminished; the sigmoid derivative never exceeds 0.25 and vanishes in the tails — the root of vanishing gradients in deep nets.

Exercise 10.1 — Forward pass by hand [Math]

Exercise

Consider a single hidden-layer network with input $X = (X_1, X_2) = (2, 3)$, $K = 2$ ReLU hidden units, and a linear output. The parameters are

$$\begin{aligned} \text{Unit 1: } w_1 &= (1, 1), w_{10} = -4, & \beta_0 &= 1, \beta_1 = 2, \beta_2 = -1. \\ \text{Unit 2: } w_2 &= (-1, 2), w_{20} = 0, \end{aligned}$$

- 1 Compute the two hidden activations A_1, A_2 using $g(z) = \max(0, z)$.
- 2 Compute the network output $\hat{f}(X)$.
- 3 Treating the output as a binary-classification logit, apply the sigmoid $\sigma(z) = 1/(1 + e^{-z})$ to obtain $\Pr(Y = 1 \mid X)$ and give the predicted class at threshold 0.5.

Solution — Exercise 10.1 (1/2)

Solution

Method. Work inside-out: each hidden unit first forms its affine pre-activation $z_k = w_{k0} + w_k^\top X$, then applies the nonlinearity $g(z) = \max(0, z)$ element-wise.

Step 1 — pre-activations.

$$z_1 = w_{10} + w_1^\top X = -4 + (1)(2) + (1)(3) = -4 + 5 = 1,$$

$$z_2 = w_{20} + w_2^\top X = 0 + (-1)(2) + (2)(3) = -2 + 6 = 4.$$

Step 2 — ReLU activations. Both pre-activations are positive, so ReLU passes them through unchanged:

$$A_1 = \max(0, 1) = 1, \quad A_2 = \max(0, 4) = 4.$$

Common mistake: forgetting the bias $w_{10} = -4$ and reporting $z_1 = 5$ — always add the bias before applying g .

Solution — Exercise 10.1 (2/2)

Solution

Step 3 — output.

$$\hat{f}(X) = \beta_0 + \beta_1 A_1 + \beta_2 A_2 = 1 + (2)(1) + (-1)(4) = -1.$$

Step 4 — sigmoid probability.

$$\sigma(-1) = \frac{1}{1 + e^1} = \frac{1}{1 + 2.718} = 0.269.$$

Since $0.269 < 0.5$, the predicted class is $Y = 0$. *Shortcut check:* σ is monotone, so $\sigma(z) \geq 0.5 \iff z \geq 0$ — the negative logit -1 already announces class 0 before any arithmetic.

Take-away. A forward pass is just alternating affine maps and element-wise nonlinearities. Note that if z_1 had been negative, ReLU would have set that unit to exactly 0, “switching it off”.

Common mistake: squashing the hidden units with the sigmoid too — here g is ReLU; only the output logit is passed through σ .

Exercise 10.2 — Why nonlinearity? [Concept]

Exercise

- 1 Show algebraically that a two-layer network with *identity* activations, $A^{(1)} = W^{(1)}X + b^{(1)}$ and $\hat{Y} = W^{(2)}A^{(1)} + b^{(2)}$, is equivalent to a single linear model. What does this imply for stacking layers without a nonlinearity?
- 2 Give two concrete advantages of ReLU over the sigmoid activation.
- 3 Explain the “vanishing gradient” problem of the sigmoid using its derivative.

Solution — Exercise 10.2 (1/2)

Solution

Part 1 — linear collapse. Substituting,

$$\hat{Y} = W^{(2)}(W^{(1)}X + b^{(1)}) + b^{(2)} = \underbrace{(W^{(2)}W^{(1)})}_{W'}X + \underbrace{(W^{(2)}b^{(1)} + b^{(2)})}_{b'} = W'X + b'.$$

This is a single affine (linear) map. Hence *any* number of linear layers collapses to one linear layer: depth buys nothing without a nonlinearity. Nonlinear g is what lets a network approximate curved decision boundaries and functions. *Common mistake: hoping the biases add flexibility — they collapse too, into the single vector b' .*

Part 2 — ReLU advantages.

- *No saturation for $z > 0$:* the derivative is exactly 1, so gradients pass through undiminished, enabling deep networks to train.
- *Cheap and sparse:* $\max(0, z)$ is a single comparison; it also sets many units to exactly 0, giving sparse, efficient representations.

Solution — Exercise 10.2 (2/2)

Solution

Part 3 — vanishing gradients. For $\sigma(z) = 1/(1 + e^{-z})$ the derivative is

$$\sigma'(z) = \sigma(z)(1 - \sigma(z)) \leq 0.25,$$

with its maximum at $z = 0$, where $\sigma'(0) = 0.5 \times 0.5 = 0.25$, and $\sigma'(z) \rightarrow 0$ as $|z| \rightarrow \infty$ (already at $z = 5$, $\sigma' \approx 0.0066$). In backpropagation the layer gradients *multiply*, so through L sigmoid layers the signal shrinks by up to 0.25^L — for $L = 10$ that is $0.25^{10} \approx 10^{-6}$: it vanishes, and early layers barely learn. ReLU's derivative of 1 on the active region avoids this.

Interpretation. Vanishing gradients are a property of the *product* of many small local derivatives, not of any one layer: each sigmoid layer is harmless alone, but ten in a row mute the error signal reaching the early weights.

Common mistake: evaluating the derivative as $\sigma'(z) = z(1 - z)$ — the formula plugs in the output $\sigma(z)$, not the pre-activation z itself.

Extended Exercise 10.1 — Tiny network by hand [Math]

Extended exercise (15 min)

A network has two inputs, one hidden layer of **two ReLU units**, and a single **sigmoid** output. Take input $x = (x_1, x_2) = (1, 2)$ and target $y = 0$, with

$$\text{Unit 1: } w_1 = (1, 1), \quad b_1 = -1; \quad \text{Unit 2: } w_2 = (0, 1), \quad b_2 = 0;$$

$$\text{logit } s = v_1 a_1 + v_2 a_2 + c, \quad v = (1, -\tfrac{1}{2}), \quad c = 0, \quad \hat{y} = \sigma(s).$$

Use squared-error loss $L = \frac{1}{2}(\hat{y} - y)^2$ and learning rate $\eta = 0.1$.

- ➊ **Forward pass:** find z_k , $a_k = \max(0, z_k)$, the logit s , $\hat{y} = \sigma(s)$ and the loss L .
- ➋ **Backprop:** write the chain rule for $\partial L / \partial w_{11}$ (the $x_1 \rightarrow$ unit-1 weight) and evaluate each factor.
- ➌ **Update:** take one gradient-descent step on w_{11} and interpret it.

Solution — Extended Exercise 10.1 (1/3)

Forward pass

Method. Alternate affine maps and nonlinearities, $z_k \rightarrow a_k \rightarrow s \rightarrow \hat{y} \rightarrow L$, and keep every intermediate value — backpropagation will reuse each one.

Hidden pre-activations and ReLU.

$$z_1 = (1)(1) + (1)(2) - 1 = 2, \quad a_1 = \max(0, 2) = 2,$$

$$z_2 = (0)(1) + (1)(2) + 0 = 2, \quad a_2 = \max(0, 2) = 2.$$

Output logit and sigmoid.

$$s = (1)(2) + (-\tfrac{1}{2})(2) + 0 = 1, \quad \hat{y} = \sigma(1) = \frac{1}{1 + e^{-1}} = \frac{1}{1 + 0.368} = 0.731.$$

Squared-error loss (target $y = 0$, so the error is $\hat{y}$ itself):

$$L = \tfrac{1}{2}(\hat{y} - y)^2 = \tfrac{1}{2}(0.731)^2 = 0.267.$$

Solution — Extended Exercise 10.1 (2/3)

Backpropagation: the chain rule

The weight w_{11} reaches the loss along $z_1 \rightarrow a_1 \rightarrow s \rightarrow \hat{y} \rightarrow L$:

$$\frac{\partial L}{\partial w_{11}} = \frac{\partial L}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial s} \frac{\partial s}{\partial a_1} \frac{\partial a_1}{\partial z_1} \frac{\partial z_1}{\partial w_{11}}.$$

Evaluate each factor:

$$\frac{\partial L}{\partial \hat{y}} = \hat{y} - y = 0.731, \quad \frac{\partial \hat{y}}{\partial s} = \sigma(s)(1 - \sigma(s)) = (0.731)(0.269) = 0.197,$$

$$\frac{\partial s}{\partial a_1} = v_1 = 1, \quad \frac{\partial a_1}{\partial z_1} = 1\{z_1 > 0\} = 1, \quad \frac{\partial z_1}{\partial w_{11}} = x_1 = 1.$$

The first two factors combine into the reusable “error signal” $\delta = (\hat{y} - y) \sigma'(s) = 0.731 \times 0.197 = 0.144$.
Common mistake: evaluating $\sigma'(s)$ by plugging $s = 1$ into $u(1 - u)$ — the u in $\sigma'(s) = u(1 - u)$ is $u = \sigma(s) = 0.731$.

Solution — Extended Exercise 10.1 (3/3)

Gradient step and interpretation

Multiply the factors.

$$\frac{\partial L}{\partial w_{11}} = 0.731 \times 0.197 \times 1 \times 1 \times 1 = 0.144.$$

Gradient-descent step ($\eta = 0.1$):

$$w_{11} \leftarrow w_{11} - \eta \frac{\partial L}{\partial w_{11}} = 1 - 0.1(0.144) = 0.986.$$

Check: redoing the forward pass with $w_{11} = 0.986$ gives $s = 0.986$, $\hat{y} = 0.728$, $L = 0.265 < 0.267$ — the loss indeed fell.

Interpretation. $\hat{y} = 0.731$ overshoots the target 0, so the gradient is positive and the step *lowers* w_{11} , pulling $\hat{y}$ down. The shared factor $\delta = (\hat{y} - y) \sigma'(s) = 0.144$ is the output “error signal”: backprop reuses it for every weight, multiplying only by that weight’s local input. Had z_1 been negative, $\partial a_1 / \partial z_1 = 0$ would kill the gradient — a “dead” ReLU unit stops learning.

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Multilayer perceptron (MLP)

Stack hidden layers:

Forward pass

$$A^{(1)} = g(W^{(1)}X + b^{(1)}),$$

$$A^{(2)} = g(W^{(2)}A^{(1)} + b^{(2)}),$$

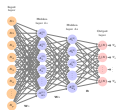
$$\vdots$$

$$\hat{Y} = h(W^{(L)}A^{(L-1)} + b^{(L)}).$$

How to read this

h is the output activation: identity for regression, softmax for multiclass classification.

MLP as a graph: stacking hidden layers



Takeaway

Each layer transforms the previous layer's features; stacking them lets the network build progressively more abstract representations from the raw inputs.

Exercise 10.3 — Counting parameters [Math]

Exercise

A fully-connected classifier has the architecture

$$\underbrace{10}_{\text{inputs}} \rightarrow 32 \rightarrow 16 \rightarrow \underbrace{3}_{\text{softmax outputs}},$$

where every hidden and output unit has a bias term.

- 1 State the general rule for the number of trainable parameters in a dense layer with n_{in} inputs and n_{out} units.
- 2 Count the parameters layer by layer.
- 3 Give the total.

Solution — Exercise 10.3 (1/2)

Solution

Step 1 — the rule and why. A dense layer maps $n_{\text{in}} \rightarrow n_{\text{out}}$: every input–output pair carries one weight (an $n_{\text{in}} \times n_{\text{out}}$ matrix), and each output unit adds one bias, so

$$\# \text{params} = n_{\text{in}} n_{\text{out}} + n_{\text{out}}.$$

The input layer itself holds *no* parameters — it only passes the feature values forward.

Step 2 — layer by layer.

$$\text{Layer 1 (10} \rightarrow \text{32)} : 10 \times 32 + 32 = 320 + 32 = 352,$$

$$\text{Layer 2 (32} \rightarrow \text{16)} : 32 \times 16 + 16 = 512 + 16 = 528,$$

$$\text{Output (16} \rightarrow \text{3)} : 16 \times 3 + 3 = 48 + 3 = 51.$$

Solution — Exercise 10.3 (2/2)

Solution

Step 3 — total.

$$352 + 528 + 51 = \boxed{931} \text{ trainable parameters.}$$

Where the parameters live. The three blocks contribute

$$352/931 \approx 38\%, \quad 528/931 \approx 57\%, \quad 51/931 \approx 5\% :$$

the $32 \rightarrow 16$ block dominates because its weight *product* $n_{\text{in}}n_{\text{out}}$ is largest, while all biases together ($32 + 16 + 3 = 51$) are almost a rounding error.

Take-away. Biases add only n_{out} each but are easy to forget. Most parameters live in the widest weight matrices, which is why hidden-layer width drives model size — and why parameter counting is a quick sanity check before training.

Common mistake: attaching parameters to the 10 input units, or dropping the $+n_{\text{out}}$ bias term — both corrupt the total.

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Why convolutions?

Image data have local, translation-invariant structure.

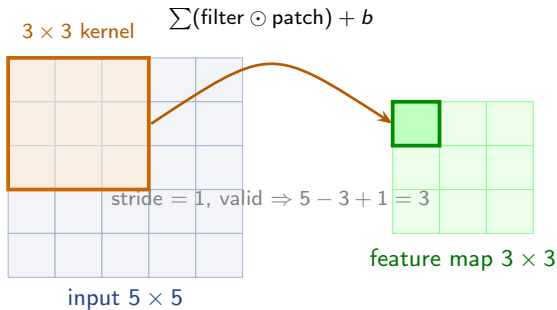
Takeaway

A cat remains a cat shifted by 10 pixels. Fully-connected networks ignore this and need much more data.

Two key layer types

- **Convolutional layers:** each unit looks at a small patch of the input; weights are shared across positions.
- **Pooling layers:** down-sample (max or average) for translation tolerance.

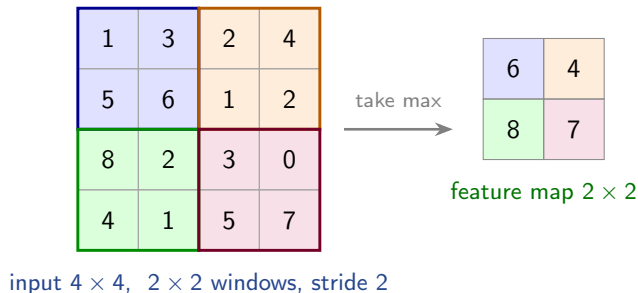
The convolution operation



Takeaway

Each output cell is a dot product of the shared-weight filter with a local input patch. Weight sharing plus locality make convolutional neural networks (CNNs) data-efficient for images.

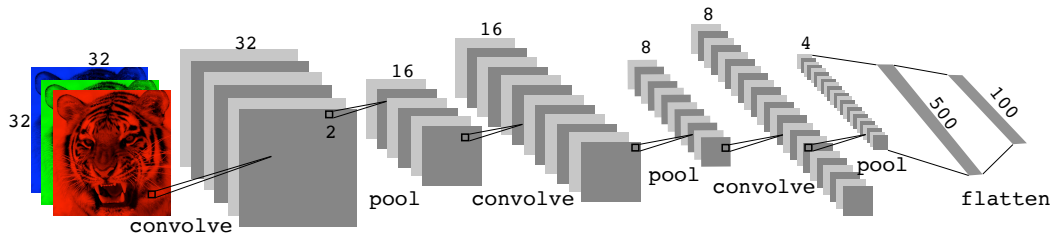
Max-pooling: downsampling by local maxima



Takeaway

Each coloured window is replaced by its maximum, shrinking H, W by the stride while keeping the depth. This buys small-shift tolerance and cuts computation — and pooling has *no* trainable parameters.

A simple CNN architecture



Takeaway

The pattern repeats: convolutions extract local features, pooling shrinks the maps, and a final dense layer classifies. Early layers learn edges; deeper layers assemble them into whole objects.

Modern CNN tricks

- **Batch normalisation:** re-centre and scale each layer's inputs per mini-batch — stabilises and speeds up training.
- **Dropout:** randomly switch off units so the net cannot rely on any single one (detailed in the Fitting section).
- **Data augmentation:** random crops, flips, colour jitter — free extra “images” that teach invariance.
- **Skip / residual connections** (ResNet): let gradients bypass layers, enabling very deep networks.
- **Transfer learning:** reuse features pretrained on ImageNet, then fine-tune on your (smaller) task.

Takeaway

These tricks turned CNNs from finicky to reliable. Most are cheap to add and mainly improve *generalisation*.

Exercise 10.4 — CNN layer arithmetic [Math]

Exercise

A colour image of size $32 \times 32 \times 3$ is fed to a convolutional layer with 16 filters, each 5×5 , stride 1, no padding (“valid”). Recall the spatial-size formula

$$O = \left\lfloor \frac{W - F + 2P}{S} \right\rfloor + 1.$$

- 1 Compute the output volume (height, width, depth) of the conv layer.
- 2 Compute the number of trainable parameters in the conv layer (include biases).
- 3 A 2×2 max-pooling layer (stride 2) follows. What is its output volume and how many parameters does it add?
- 4 Redo part (1) if instead padding $P = 2$ (“same”) is used.

Solution — Exercise 10.4 (1/2)

Solution

Part 1 — conv output. With $W = 32$, $F = 5$, $P = 0$, $S = 1$:

$$O = \frac{32 - 5 + 0}{1} + 1 = 28.$$

Each filter slides over the image and produces *one* 28×28 feature map, so depth = number of filters: the output volume is $28 \times 28 \times 16$.

Part 2 — conv parameters. Each filter spans the full input depth: $5 \times 5 \times 3 = 75$ weights, $+1$ bias = 76 per filter. With 16 filters:

$$16 \times 76 = \boxed{1216} \text{ parameters.}$$

Why so few? A dense layer from the $32 \times 32 \times 3 = 3072$ inputs to the same $28 \times 28 \times 16 = 12544$ outputs would need $3072 \times 12544 + 12544 \approx 38.5$ million parameters — weight sharing cuts that by a factor of about 30 000.

Solution — Exercise 10.4 (2/2)

Solution

Part 3 — pooling. Max-pooling with $F = 2$, $S = 2$:

$$O = \frac{28 - 2}{2} + 1 = 14,$$

giving $14 \times 14 \times 16$: pooling acts on each of the 16 channels separately, so depth is preserved. It has **no** trainable parameters — it only takes a maximum.

Part 4 — with padding $P = 2$.

$$O = \frac{32 - 5 + 2(2)}{1} + 1 = \frac{31}{1} + 1 = 32.$$

Output $32 \times 32 \times 16$: “same” padding preserves the spatial size. In general, stride-1 “same” padding takes $P = (F - 1)/2$ — here $(5 - 1)/2 = 2$ — so conv layers can be stacked without shrinking the maps.

Common mistake: adding a $+2P$ term for the pooling layer (it uses $P = 0$) or forgetting to divide by the stride before adding 1.

Extended Exercise 10.2 — CNN architecture arithmetic [Math]

Extended exercise (15 min)

A $64 \times 64 \times 3$ RGB image passes through the stack below (convolutions use the padding shown; pooling is max-pooling, no padding). Recall $O = \lfloor (W - F + 2P)/S \rfloor + 1$.

Layer	Kernel	Stride / Pad	Filters
Conv1	3×3	1 / 1	32
Pool1	2×2	2 / 0	—
Conv2	3×3	1 / 1	64
Pool2	2×2	2 / 0	—
Dense	softmax	—	10

- 1 Give the output volume ($H \times W \times \text{depth}$) after each layer.
- 2 Count the learnable parameters of every layer, and the total.
- 3 What input region does one unit of the last conv map see (receptive field)?

Solution — Extended Exercise 10.2 (1/3)

Spatial dimensions

Apply $O = (W - F + 2P)/S + 1$ to H and W ; depth = number of filters (pooling keeps depth).

$$\text{Conv1 : } (64 - 3 + 2)/1 + 1 = 64 \quad \Rightarrow \quad 64 \times 64 \times 32,$$

$$\text{Pool1 : } (64 - 2)/2 + 1 = 32 \quad \Rightarrow \quad 32 \times 32 \times 32,$$

$$\text{Conv2 : } (32 - 3 + 2)/1 + 1 = 32 \quad \Rightarrow \quad 32 \times 32 \times 64,$$

$$\text{Pool2 : } (32 - 2)/2 + 1 = 16 \quad \Rightarrow \quad 16 \times 16 \times 64.$$

Flattening the last volume gives $16 \times 16 \times 64 = 16384$ features into the dense layer. “Same” padding preserves H, W ; each pool halves them.

Common mistake: flattening only $16 \times 16 = 256$ features — the depth counts too: $16 \times 16 \times 64 = 16\,384$.

Solution — Extended Exercise 10.2 (2/3)

Learnable parameters

A conv filter spans the *full input depth*, so $\# = (F^2 d_{\text{in}} + 1) \times (\# \text{filters})$; pooling has none.

$$\text{Conv1} : (3^2 \cdot 3 + 1) \times 32 = 28 \times 32 = 896,$$

$$\text{Conv2} : (3^2 \cdot 32 + 1) \times 64 = 289 \times 64 = 18\,496,$$

$$\text{Dense} : 16384 \times 10 + 10 = 163\,850,$$

$$\text{Pool1, Pool2} : 0.$$

$$\text{Total} = 896 + 18\,496 + 163\,850 = \boxed{183\,242} \text{ parameters.}$$

Solution — Extended Exercise 10.2 (3/3)

Receptive field and interpretation

Method. Track the receptive field r and the jump j (input pixels between neighbouring units): a layer with filter F and stride S updates $r \leftarrow r + (F - 1)j$, then $j \leftarrow jS$. Start at $r = 1, j = 1$:

$$\text{Conv1 } (F=3, S=1) : r = 1 + 2 \times 1 = 3, \quad j = 1,$$

$$\text{Pool1 } (F=2, S=2) : r = 3 + 1 \times 1 = 4, \quad j = 2,$$

$$\text{Conv2 } (F=3, S=1) : r = 4 + 2 \times 2 = 8, \quad j = 2,$$

$$\text{Pool2 } (F=2, S=2) : r = 8 + 1 \times 2 = 10, \quad j = 4.$$

One unit of the final map therefore sees a 10×10 patch of the original image. **Take-aways.** (i) Convolutions are parameter-thrifty: the two conv layers use $< 20\text{k}$ weights through *weight sharing*, while the single dense layer holds 164k — about 89% of the model. (ii) Depth plus pooling grow the receptive field, letting deep units assemble local edges into global shapes.

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Recurrent neural networks (RNNs)

Process sequences $x_1, \dots, x_T$ by carrying a hidden state forward:

$$h_t = g(W_x x_t + W_h h_{t-1} + b).$$

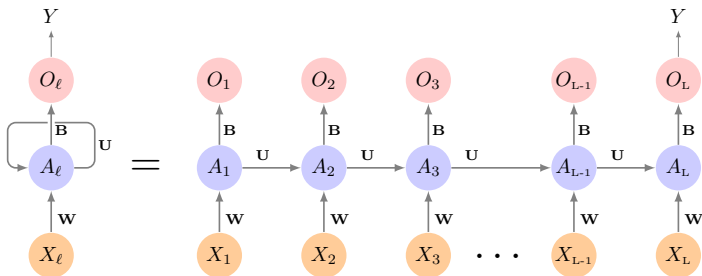
Reading the recurrence

- x_t : input at step t ; h_t : hidden state (“memory”) at step t .
- W_x : weights on the new input; W_h : weights on the previous state h_{t-1} ; b : bias.
- The *same* W_x, W_h, b are reused at every step — weight sharing across time.

Takeaway

Suitable for time series, text, audio. Plain RNNs suffer from vanishing/exploding gradients; LSTM and GRU add gating to carry long-range information.

An RNN unrolled in time



Takeaway

Unrolling shows the recurrence as a deep feed-forward net that reuses the same weights at every step; training uses backpropagation “through time”.

Transformers (briefly)

Takeaway

“Attention is all you need” (Vaswani et al., 2017). Self-attention computes weighted combinations of all positions in a sequence. Beats RNNs on most NLP tasks; foundation of modern LLMs.

Self-attention in one line

Each position forms a *query* and compares it against every position's *key*; the resulting weights average the *values*. So every output token can look directly at every input token — no matter how far apart, and all positions are processed in parallel (unlike an RNN's step-by-step loop).

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Loss and optimisation

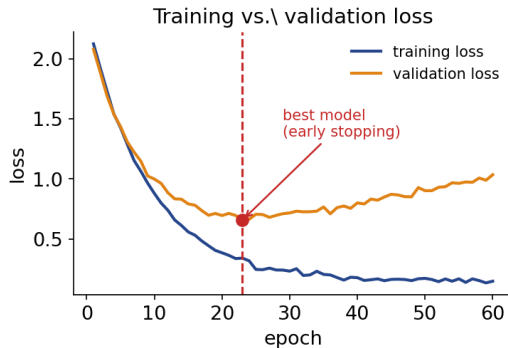
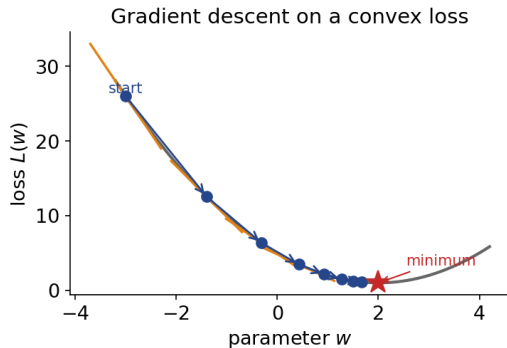
Choose a loss $L(\theta) = \sum_i \ell(y_i, \hat{y}_i; \theta)$ and minimise by stochastic gradient descent:

$$\theta \leftarrow \theta - \eta \nabla_{\theta} \ell_{\text{minibatch}}(\theta).$$

Reading the update

- θ : all weights and biases; $\nabla_{\theta} \ell$: gradient (*steepest-ascent* direction) of the loss.
- Step *against* the gradient, scaled by the learning rate η (too large diverges, too small crawls — see Exercise 10.5).
- “Stochastic”: each step uses one random mini-batch (32/64/128) — a cheap, noisy estimate of the full-data gradient.
- Modern optimisers add memory: SGD+momentum, Adam, RMSprop.

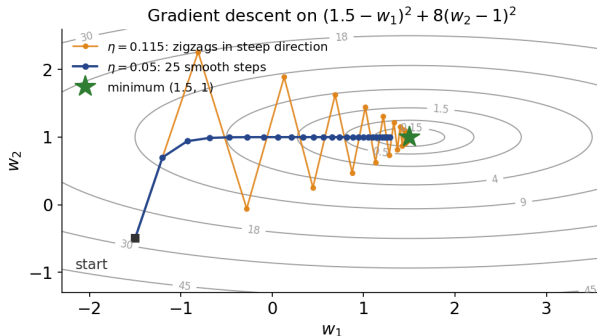
Training a network: descent and early stopping



Takeaway

Gradient descent steps downhill along the loss surface (left). In practice, watch training *and* validation loss and stop at the validation minimum, before overfitting sets in (right).

Gradient descent on a loss surface: the real path



Takeaway

Each step is perpendicular to the local contour: with $\eta = 0.05$ the loss falls from 27 to 0.05 in 25 steps — fast down the steep w_2 axis, then a crawl along the flat valley — while the too-large $\eta = 0.115$ overshoots and zigzags in the steep direction.

Backpropagation in one slide

The idea in two passes

- **Forward pass:** push x through the layers, caching each activation.
- **Backward pass:** send an “error signal” δ from the loss back to every weight, multiplying local derivatives (the chain rule).
- Each weight’s gradient = (error signal at its output) $\times$ (its input); shared factors are reused, so the cost is $\approx$ one forward pass.

Takeaway

Backprop is just the chain rule applied from output back to input — reverse-mode automatic differentiation. Implemented in PyTorch, JAX, TensorFlow, so you rarely write it by hand.

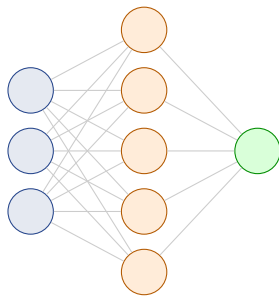
Regularisation in deep learning

Modern networks can have more parameters than data points but still generalise. Why?

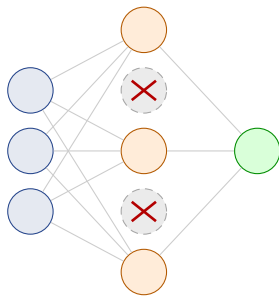
How to read this

- Implicit regularisation by SGD.
- Dropout, weight decay (ℓ_2), early stopping.
- Data augmentation.
- Architecture choice is itself a strong inductive bias.

Dropout: randomly switch off units during training



Full network

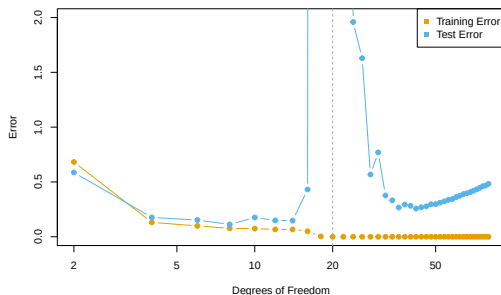


During training: 2 units dropped

Takeaway

On each mini-batch, every hidden unit is kept with probability $1 - \rho$ and otherwise *zeroed*, so the net cannot lean on any single unit and learns redundant, robust features. At test time all units are used (weights rescaled).

Double descent



Why the second dip

Test error first traces the classical U-shape, then peaks at the *interpolation threshold* (the model *just* fits the data and is unstable), and falls again as extra parameters let SGD pick a smoother interpolant.

Exercise 10.5 — One gradient-descent step [Math]

Exercise

Consider minimising the loss $L(w) = (w - 3)^2$ by gradient descent, $w \leftarrow w - \eta L'(w)$, starting from $w = 2$.

- 1 Write $L'(w)$ and evaluate it at $w = 2$.
- 2 Perform one update with learning rate $\eta = 0.1$.
- 3 Show that the update rule for the *error* $e = w - 3$ is $e \leftarrow (1 - 2\eta) e$, and deduce the range of η for which gradient descent converges.
- 4 What happens with $\eta = 1.5$? Interpret.

Solution — Exercise 10.5 (1/2)

Solution

Part 1 — gradient. $L'(w) = 2(w - 3)$, so $L'(2) = 2(2 - 3) = -2$.

Part 2 — one step ($\eta = 0.1$).

$$w \leftarrow 2 - 0.1 \times (-2) = 2 + 0.2 = 2.2.$$

We moved from 2 toward the minimiser $w^* = 3$: good.

Part 3 — convergence range. Substituting $L'(w) = 2(w - 3)$ into the update,

$$w \leftarrow w - \eta 2(w - 3) \implies (w - 3) \leftarrow (w - 3) - 2\eta(w - 3) = (1 - 2\eta)(w - 3).$$

So $e_{t+1} = (1 - 2\eta) e_t$, hence $e_t = (1 - 2\eta)^t e_0$: a geometric sequence, which dies out iff its ratio satisfies $|1 - 2\eta| < 1$, i.e.

$$0 < \eta < 1.$$

With $\eta = 0.1$ the ratio is 0.8 — each step removes 20% of the remaining error.

Solution — Exercise 10.5 (2/2)

Solution

Part 4 — $\eta = 1.5$. Then $1 - 2\eta = -2$, so $|1 - 2\eta| = 2 > 1$: the error *doubles* each step and flips sign. Concretely $w \leftarrow 2 - 1.5(-2) = 5$; now $|5 - 3| = 2 > |2 - 3| = 1$, and iterating,

t	0	1	2	3
w_t	2	5	-1	11
$e_t = w_t - 3$	-1	2	-4	8

The steps overshoot the minimum ever more wildly: gradient descent **diverges**. This is the classic “learning rate too large” failure — η trades off speed against stability.

Interpretation. The extremes are instructive: $\eta = 0.5$ makes the ratio $1 - 2\eta = 0$ and lands on $w^* = 3$ in *one* step (possible only because this loss is exactly quadratic), while a tiny η converges but crawls.

Common mistake: concluding “smaller is always safer, so smaller is better” — at $\eta = 0.01$ the ratio is 0.98 and hundreds of steps are needed here.

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Python lab — run it live in the notebook

Companion notebook: `chapter_10_lab.ipynb`

The code in this section is meant to be **demonstrated live**. Open the companion Jupyter notebook and run it cell by cell:

- building a small multilayer perceptron in **PyTorch**;
- the data pipeline, model definition, and training loop;
- evaluating and tuning the network.

Requires `torch`; the notebook ends with hands-on exercises.

An MLP in PyTorch

```
import torch
from torch import nn, optim

class MLP(nn.Module): # a model = a subclass of nn.Module
    def __init__(self, p, k=64, K=2): # p inputs, k hidden units, K classes
        super().__init__()
        self.net = nn.Sequential(# layers applied in order:
            nn.Linear(p, k), # affine map  $Wx + b$  ( $p \rightarrow k$ )
            nn.ReLU(), # nonlinearity  $\max(0, z)$ 
            nn.Dropout(0.2), # randomly zero 20% of units (regularises)
            nn.Linear(k, k), nn.ReLU(), nn.Dropout(0.2), # second hidden layer
            nn.Linear(k, K)) # output layer: K logits (no softmax here)
    def forward(self, x): # forward pass: how inputs become outputs
        return self.net(x)

model = MLP(p=Xtr.shape[1]) # build the network for our p predictors
opt = optim.Adam(model.parameters(), lr=1e-3) # optimiser: Adam, learning rate 0.001
```

Exercise 10.6 — Softmax and cross-entropy [Python]

Exercise

A 3-class classifier outputs the logits $z = (2.0, 1.0, 0.1)$ for one observation whose true label is class 0.

- 1 By hand, compute the softmax probabilities $p_k = e^{z_k} / \sum_j e^{z_j}$.
- 2 Compute the cross-entropy loss $\ell = -\log p_y$ for the true class $y = 0$.
- 3 Write short NumPy code that reproduces both and prints them.

Solution — Exercise 10.6 (1/3)

Solution

Method. Softmax exponentiates each logit (making it positive) and divides by the sum, turning arbitrary scores into a probability vector; cross-entropy then scores only the probability given to the *true* class.

Step 1 — exponentiate.

$$e^{2.0} = 7.389, \quad e^{1.0} = 2.718, \quad e^{0.1} = 1.105, \quad \sum = 11.212.$$

Step 2 — softmax probabilities.

$$p = \left(\frac{7.389}{11.212}, \frac{2.718}{11.212}, \frac{1.105}{11.212} \right) = (0.659, 0.242, 0.099).$$

Check: the three probabilities sum to 1.000.

Step 3 — cross-entropy for $y = 0$.

$$\ell = -\log p_0 = -\log(0.659) = 0.417.$$

Common mistake: using $\log_{10}$ instead of the natural log ($-\ln 0.659 = 0.417$).

Solution — Exercise 10.6 (2/3)

Solution

```
import numpy as np # numerics library

z = np.array([2.0, 1.0, 0.1]) # logits from the network
p = np.exp(z) / np.exp(z).sum() # softmax: exponentiate, normalise
print("probs:", np.round(p, 3)) # [0.659 0.242 0.099]

y = 0 # true class label
loss = -np.log(p[y]) # cross-entropy = -log(true-class prob)
print("loss:", round(float(loss), 3)) # 0.417
```

Solution — Exercise 10.6 (3/3)

Interpretation

The model assigns the highest probability (0.659) to the correct class, so the loss is modest. A perfect prediction ($p_0 = 1$) gives loss 0; a confidently wrong prediction sends $-\log p_0 \rightarrow \infty$ — cross-entropy punishes confident errors hardest, exactly the gradient signal a classifier needs.

Why the “log-sum-exp” trick works. Softmax is shift-invariant: dividing top and bottom by e^c gives $p_k = e^{z_k - c} / \sum_j e^{z_j - c}$, unchanged for any c . Choosing $c = \max_k z_k = 2.0$ here,

$$e^0 = 1, \quad e^{-1} = 0.368, \quad e^{-1.9} = 0.150, \quad p = \frac{1}{1.518} (1, 0.368, 0.150) = (0.659, 0.242, 0.099),$$

the same probabilities — but every exponent is now ≤ 0 , so nothing can overflow no matter how large the logits get.

Common mistake: applying softmax yourself and then calling PyTorch's `nn.CrossEntropyLoss` — that loss already includes the softmax and expects raw logits; softmaxing twice silently distorts the loss.

Extended Exercise 10.3 — Overfitting and regularisation [Python]

Extended exercise (15 min)

Fit a **PyTorch** multilayer perceptron to the `Default` data, *watch it overfit*, then tame it. Encode student as 0/1, standardise (`balance`, `income`, `student`), and predict `default`.

- 1 Train on a *small* subset ($n_{\text{train}} = 200$) with a wide MLP (two hidden layers of 128, no dropout) and no weight decay, recording the training and validation cross-entropy after every epoch.
- 2 Where does the validation loss turn upward, and what does the training loss do?
- 3 Refit with strong L2 regularisation (`weight_decay=0.1` in Adam); compare the validation loss and say what *early stopping* would choose.

Solution — Extended Exercise 10.3 (1/3)

Code: load / standardise / convert to tensors

```
import pandas as pd, torch
from torch import nn, optim
from sklearn.preprocessing import StandardScaler
from sklearn.model_selection import train_test_split

df = pd.read_csv("Default.csv")
df["y"] = (df["default"] == "Yes").astype(int) # response -> 0/1
df["student"] = (df["student"] == "Yes").astype(int) # dummy-code student
X = df[["balance", "income", "student"]].astype(float).values
Xtr, Xte, ytr, yte = train_test_split(X, df["y"].values,
                                     train_size=200, random_state=0, stratify=df["y"]) # tiny train set!
sc = StandardScaler().fit(Xtr) # scale using TRAIN stats only
Xtr, Xte = sc.transform(Xtr), sc.transform(Xte)
# PyTorch works on tensors: features float32, labels int64
Xtr, Xte = map(lambda a: torch.tensor(a, dtype=torch.float32), (Xtr, Xte))
ytr, yte = map(lambda a: torch.tensor(a, dtype=torch.int64), (ytr, yte))
```

Solution — Extended Exercise 10.3 (2/3)

Code: train and record the loss after every epoch

```
def curve(weight_decay, epochs=400):
    torch.manual_seed(1) # reproducible weights
    net = nn.Sequential(nn.Linear(3,128), nn.ReLU(), # wide MLP
                        nn.Linear(128,128), nn.ReLU(), nn.Linear(128,2))
    loss_fn = nn.CrossEntropyLoss() # softmax + log-loss in one
    opt = optim.Adam(net.parameters(), lr=1e-2,
                     weight_decay=weight_decay) # L2 penalty lives here
    tr, va = [], []
    for _ in range(epochs): # 1 epoch = 1 full-batch step here
        opt.zero_grad() # reset gradients
        loss = loss_fn(net(Xtr), ytr) # forward pass on TRAIN
        loss.backward(); opt.step() # backprop + update
        with torch.no_grad(): # eval: no gradients
            tr.append(loss.item()); va.append(loss_fn(net(Xte), yte).item())
    return tr, va
tr0, va0 = curve(0.0) # unregularised -> memorises the 200 points
tr1, va1 = curve(0.1) # strong weight decay (L2)
```

Solution — Extended Exercise 10.3 (3/3)

Typical result and interpretation

Typical values (exact numbers vary with seed and version):

unreg: train $\rightarrow$ 0.00 val-min $\approx$ 0.10 (early epochs) val-final $\approx$ 0.4-0.5
decay: train $\approx$ 0.10 val *lower and flat* $\approx$ 0.09

- *Unregularised*: training loss dives toward 0 (the net **memorises** the 200 points), but validation loss bottoms out within the first few dozen epochs and then **climbs** several-fold — textbook overfitting.
- *Weight decay*: training loss stays higher, yet validation loss is *lower and flat*: the L2 penalty trades a little fit for much better generalisation.
- *Early stopping* halts at the validation minimum — a free regulariser needing no extra penalty term.

Same recipe as the companion notebook (chapter_10_lab.ipynb); there the MLP adds dropout and mini-batches via DataLoader.

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Chapter 10 in one slide

What we accomplished

Surveyed the modern deep-learning landscape and connected it to the classical methods of the earlier chapters.

- Single- and multilayer perceptrons.
- Convolutional networks for images.
- Recurrent networks and transformers.
- Optimisation: SGD, momentum, Adam, backpropagation.
- Regularisation: dropout, weight decay, early stopping.

Key formulas at a glance

Neuron $a = g(\sum_j w_j x_j + b)$, activation g .

Single hidden layer $f(x) = \beta_0 + \sum_{k=1}^K \beta_k g(w_{k0} + \sum_j w_{kj} x_j)$.

Activations ReLU $g(z) = \max(0, z)$; sigmoid $\sigma(z) = \frac{1}{1 + e^{-z}}$; softmax $\frac{e^{z_m}}{\sum_{\ell} e^{z_{\ell}}}$.

Loss regression $\sum_i (y_i - f(x_i))^2$; classification $-\sum_i \sum_k y_{ik} \log f_k(x_i)$.

Gradient descent $\theta \leftarrow \theta - \eta \nabla_{\theta} L(\theta)$ (learning rate η).

Conv output size = $\lfloor (W - F + 2P)/S \rfloor + 1$ (input W , filter F , pad P , stride S).

Vocabulary checklist

Term	Meaning
Activation function	Element-wise nonlinearity (ReLU, tanh, sigmoid)
Hidden unit	One node in a hidden layer
Backpropagation	Reverse-mode automatic differentiation
SGD / momentum	Stochastic gradient + velocity term
Adam	Adaptive per-parameter learning rates
Dropout	Random unit muting during training
Weight decay	ℓ_2 penalty on weights
CNN	Conv + pool + dense layers for images
RNN / LSTM	Sequential models with hidden state
Transformer	Self-attention based sequence model
Double descent	Test-error second dip

Decision rules of thumb

- Tabular data ($n < 10^5$, $p < 10^3$): try gradient boosting first.
- Images: pretrained CNN (ResNet, EfficientNet) and fine-tune.
- Text: pretrained transformer; fine-tune only the head when n is small.
- Always normalise inputs and labels.
- Track training and validation loss together.
- Save the best-validation checkpoint.

Common pitfalls

Don't

- ❶ Compare a deep network to a poorly tuned baseline.
- ❷ Forget to shuffle training data before each epoch.
- ❸ Use too high a learning rate (loss explodes).
- ❹ Trust accuracy without checking calibration.
- ❺ Skip data augmentation for image / audio tasks.
- ❻ Re-use the validation set for many tuning rounds.

What's next

Chapter 13: *Multiple Testing*.

- The multiple-comparisons problem and the family-wise error rate.
- Bonferroni and Holm corrections.
- The false discovery rate and Benjamini-Hochberg.